Lab 4

1. Write a Java program to accept an integer value from the user and convert it into a byte. Display the result and observe the effect when the entered value exceeds the byte range.
2. Write a Java program to create a class Box with:

    - Instance variables: length, breadth, height
    - Constructors:
        - Default constructor
        - Parameterized constructor
    - Method: volume()

    Create objects using both constructors and display the volume.

3. Write a Java program to create a class Dog with:

    - Instance variables: name, age
    - Method: display() to print details
    - Initialize the object using:
        - Reference variable
        - Method
        - Constructor

    Display the values for each initialization technique.

4. Write a Java program to accept a double value from the user and convert it into an integer using explicit type casting. Display both values.
5. Write a Java program to accept two numbers from the user: one integer and one double. Perform addition and display the result after implicit type conversion.
6. Write a Java program to demonstrate type promotion by multiplying two byte variables and storing the result in an integer variable. Display the result.
7. Write a Java program to accept a character from the user, convert it into its corresponding integer value, and display the result.
8. Write a Java program to create a class Student with:

    - Instance variables: name, marks
    - Static variable: collegeName
    - Methods:
        - input()
        - display()

    Create **two objects** and demonstrate that the static variable is shared among all objects.